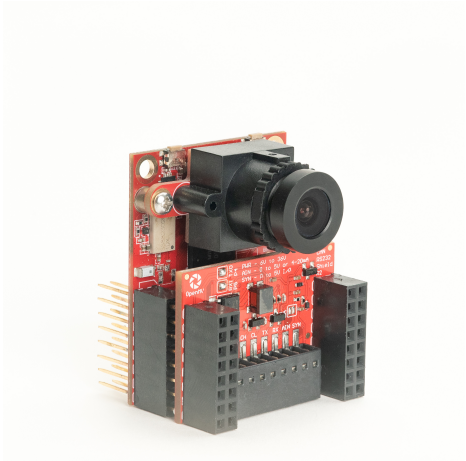
OpenMV® CAN/RS232 Shield

Datasheet

SKU: OMV-SHIELD-CAN-RS232

Description

The CAN/RS232 Shield gives your OpenMV Cam the ability to transmit RS232 signals to equipment and communicate over the CAN bus at the same time. You can transmit data up to 1 Mb/s via RS232 and 8 Mb/s via CAN (requires CAN-FD support; normal CAN maxes at 1 Mb/s). RS232 uses P4 (TX) and P5 (RX). CAN uses P1 (TX) and P3 (RX), with optional P2 for TX. Includes 6\u201336 V wide-range input, robust ADC, and robust trigger I/O.



Target Areas

Industrial communication, automotive, equipment integration, factory automation, machine interfacing, PLC connectivity.

Key Features

- **Simultaneous CAN + RS232** — CAN bus up to 8 Mb/s (CAN-FD) or 1 Mb/s (standard). RS232 up to 1 Mb/s. Both active simultaneously.
- **6–36 V wide-range input** — Reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.
- **Robust ADC input** — 0–5 V on P6 with ±36 V overvoltage protection. Optional 4–20 mA mode.
- **Robust trigger I/O** — 0–5 V on P10 via SYN pin. Tolerant to -12 V, shorts to GND and 12 V.
- **Configurable pins** — All I/O pins disconnectable via solder bridges.

Contents

1 Features

- 1.1 Communication
- 1.2 Power Input
- 1.3 ADC Input
- 1.4 Digital I/O
- 1.5 Compatibility

2 General Info

3 Company Information

4 Reference Documentation

5 Revision History

1 Features

1.1 Communication

Component	Details
RS232	RS232 transceiver on P4 (TX) and P5 (RX). Up to 1 Mb/s. Disconnectable.
CAN Bus	CAN transceiver on P1 (TX) and P3 (RX). TX optionally movable to P2. Up to 8 Mb/s with CAN-FD, 1 Mb/s standard CAN. Disconnectable.

1.2 Power Input

Component	Details
Wide-Range Input	6–36 V with reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.

1.3 ADC Input

Component	Details
AIN	Robust 0–5 V ADC input on P6 (0–3.3 V output). Overvoltage protection to ±36 V. Optional 4–20 mA mode via 240 Ω solder bridge. Disconnectable.

1.4 Digital I/O

Component	Details
SYN / P10	Robust 0–5 V digital I/O. Tolerant to -12 V, shorts to GND and 12 V. Default: open-drain input. Configurable as 0–5 V output. Disconnectable.

1.5 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	10 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/can-rs232-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.